

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination 2023

Course: Second Year B. Tech.

Branch: Civil Engineering

Subject Name: Hydraulics - I

Subject Code: BTCVC304

Max Marks: 60

Date: 18/08/2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|---|----------|----|
| A) Explain the following fluid properties: (i) Weight density, (ii) Specific gravity, (iii) viscosity. | Remember | 06 |
| B) Enlist different types of manometers. Explain Differential U-Tube Manometer with neat sketch. | CO1 | 06 |
| C) A plate 0.0254 mm distant from a fixed plate, moves at 61 cm/sec and requires a force of 0.2 kg(f)/m ² to maintain this speed. Determine the dynamic viscosity of the fluid between the plates. | CO2 | 06 |

Q. 2 Solve Any Two of the following.

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|---|----------|----|
| A) Derive a three dimensional general continuity equation in Cartesian co-ordinates. | Remember | 06 |
| B) Enlist the methods of drawing flow nets. Describe the uses and limitations of flow net. | CO2 | 06 |
| C) The stream function for a two-dimensional flow is given by $\Psi = 2xy$, calculate the velocity at the point P (2,3). Find the velocity potential function Φ . | CO 2 | 06 |

Q. 3 Solve Any Two of the following.

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|---|----------|----|
| A) Derive an expression for velocity distribution for laminar flow between two fixed parallel plates. | Remember | 06 |
| B) What is Prandtl mixing length theory? Find an expression for shear stress due to Prandtl. | CO2 | 06 |
| C) Write a short note on Boundary Layer Theory. | CO2 | 06 |

Q. 4 Solve Any Two of the following.

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|--|------|----|
| A) Write short note on non-dimensional numbers – (i) Froude number, (ii) Weber Number, (iii) Reynold's Number, (iv) Euler Number, (v) Mach Number. | CO 3 | 06 |
| B) What is meant by geometric, kinematic and dynamic similarities? | CO 4 | 06 |

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|----|---|------|----|
| C) | Explain concept of Rayleigh's method and find an expression for the drag force using Rayleigh's method on smooth sphere of diameter D , moving with uniform velocity V in a fluid of density ρ and dynamic viscosity μ . | CO 4 | 06 |
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Q. 5 Solve Any Two of the following.

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|----|--|----------|----|
| A) | What is Siphon? Explain its working with neat sketch. | Remember | 06 |
| B) | What is a surge tank and what are its functions? Describe with neat sketches different types of surge tanks. | CO 4 | 06 |
| C) | Find the diameter of a pipe of length 2000 m when the rate of flow of water through the pipe is 200 liters/s and head lost due to friction is 4 m. Take the value of $C = 50$ in <i>Chezy's formulae</i> . | CO 3 | 06 |

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